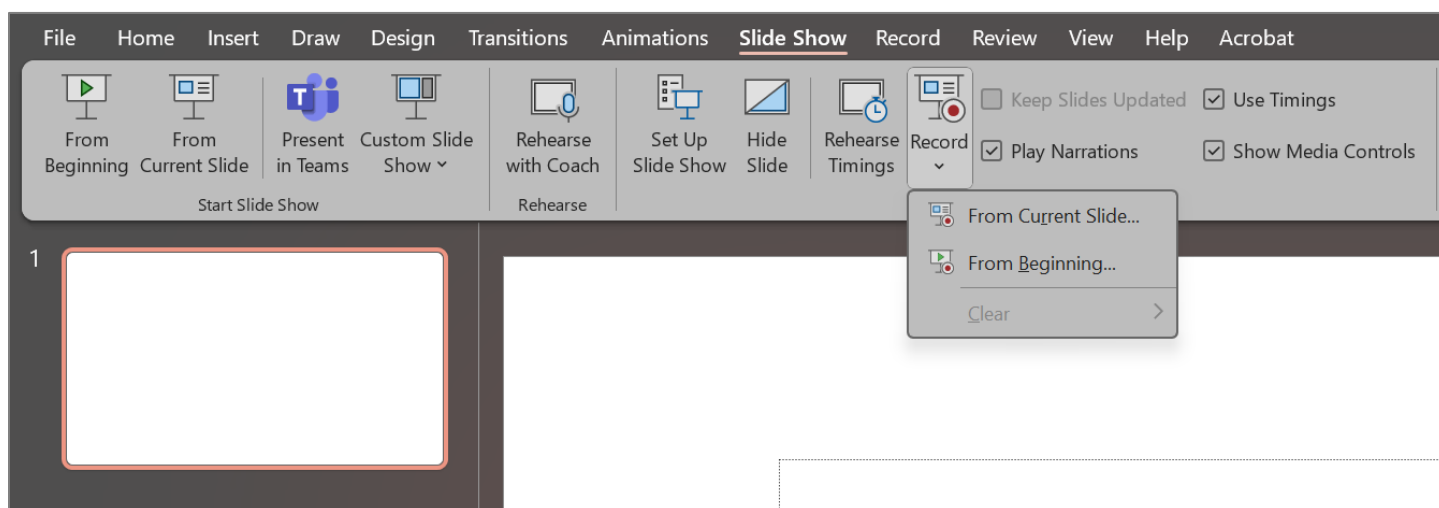


TPJ Requirements

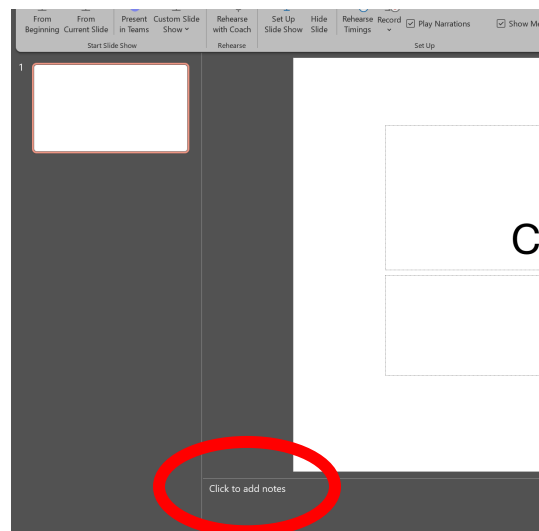
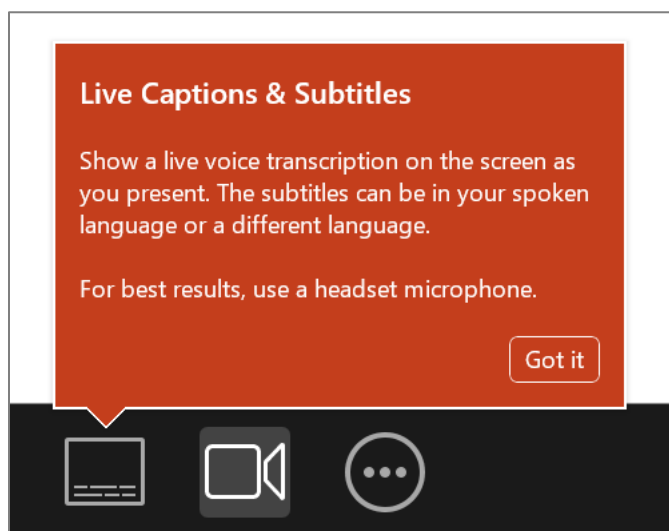
Use the comprehensive, technical, presentation skills that you have learned in previous courses (ICO155, COM101, TEC400, etc.) to help you produce your own Presentation this semester.

Your 25- to 35- minute fully polished presentation will be created by your team virtually using MS PowerPoint, a microphone, and a camera. The final PPTX will be exported as a MP4 video file and both files will be submitted for marking.

Your team members must share and alternate the speaking points throughout the presentation. Each slide will have an associated audio file autogenerated when you record your presentation.



Turn on the transcription feature so that your voices are captured in captions that can be turned on or off. Use the Notes box at the bottom of each slide to help write your speech in each slide.



Regarding GenAI: The suspected use of AI to write any portion of your slides will result in “zero” for written communications. Cite Copilot for generating ideas but use your own words.

The following slide information will appear in your presentation (*do not rename starred items):

- §01. Project title page
- §02. Introduction and research
- §03. Project description
- §04. Project images - inside and outside the final build
- §05. Complete system block diagram
- §06. Complete schematic or Graftet diagrams
- §07. PCB design diagrams
- §08. Mechanical design diagrams
- §09. *Project Hardware
- §10. *Project Software
- §11. Technical discussion on engineering problems encountered and how addressed
- §12. Complete demonstration video (max. 5 mins)
- §13. *Challenges: Lessons Learned
- §14. *Future Changes and Improvements
- §15. References / Images cited

Filming Advice

Your presentation video will not contain: an incomplete project, a breadboard, or any cardboard/tape/hot glue/similarly unfinished parts. The project in your video will appear tidy, complete, with inputs and outputs securely attached, and your soldered PCB must appear clearly in the video at some point. In the demonstration portion, consider showing the different segments of the project quickly before showing it working all together.

The depiction of your finished project in this video assignment must match your project depicted in your final Technical Report and Demonstration.

You may use any software program to **add visual effects** and overlay text elements in appropriate places along the film’s timeline, for example with titles, names, or positions. The camera used to film the demonstration should remain **steady** and filmed in a **horizontal** orientation. (Make the video wider than it is tall.) Try to use a tripod or stabilizer for the camera whenever possible. Remove all watermarks from your final video.

Choose appropriate **lighting** for each of your scenes. For example, an LED turning on or an OLED screen message might be difficult to film under bright lighting. Test the lighting and settings in several practice videos to see what works best for your project.

Try to eliminate background noise, unless it is helpful to the story. Speak **clearly** into the recording microphone, avoiding pops and hisses. Slow down your speech so the viewer can understand you clearly. Tell an exciting story! Don’t talk in a monotone. The idea of your infomercial is to provide an **entertaining** perspective on the purpose of your project.

Some of the Information on the Slides

Start with a title page, include partner names, course code & lab section, team number, title of project, a clear picture of the final project, and the professor's name. Do you immediately understand what your project does? How will you capture the audience's attention?

Order - Arrange your presentation slides in the best way that suits the flow of your discussions.

References / Images Cited - All media that appears in your Presentation, whether video, audio, or text, should be referenced on the final slides and cited appropriately. Use MLA formatting. Use the original URL, not "Google.com". Links can shorten by cutting out text after a "?" mark.

Images - All photos must be taken by the team. All diagrams must be drawn by the team using CAD software. All screenshots must be taken with a snipping tool (not a photo of the screen). Schematic should show all electrical parts of the system, but may span several images if needed. PCB should show top and bottom of design as well as top and bottom of final soldered board. Include all mechanical designs, including anything that was 3D printed.

Introduction - Do you have a Hook, introductory story, or relevant video of the practical problem your team would like to address? What was the identifiable problem or need in society that your project now addresses? Look back to your Team Proposal for the Why behind your journey.

Complete demonstration video - Insert a video demonstration of your project working from start to finish. Your project must react to all input data, process it in real-time, and demonstrate its various output results, and screencast the IoT dashboard. Screencast or record your screen while discussing code (don't take video of the screen with your camera). Max: 5-minutes.

Project Software - List all software programs and applications used to develop the code. Explain the IoT elements used in your project, their protocols, and how they are all interconnected. Use screencasting software like Camtasia, or PowerPoint itself (<https://youtu.be/aXXsIqQDsCQ>).

Project Hardware - List the power source (battery or hard-wired), all inputs, all outputs, related components, microcontroller(s), PLC(s), and the reason for their presence. Explain the technical design considerations of your project and the reason why you chose certain components.

Example: With an ultrasonic input sensor, in 10 seconds, provide its name, manufacturing part name/code, identify which ports it connects to, its general purpose, and its purpose in your project (eg. "measures this distance in these units with this precision").

Technical discussion on problems encountered and how addressed - Pick one to three (one if long, three if short) examples of engineering problems your group faced during the making of your project. Explain each detailed problem, the fix, if it created another problem, that fix, and so on until you reached a final workable solution. Regular debugging doesn't count here.

Challenges - Describe 3-5 lessons learned or challenges overcome that were most influential or memorable to your team while working on this project. Include pieces of useful advice that you would impart to the next semester students in TPJ that would help them with their projects.

Improvements - Describe in detail at least 2 significant improvements you would like to make to your project if your team was given 6 more weeks to work on it. Include a diagram as an aid.

Conclusions - How accurate is your project? Don't just say it was a success - prove why it was.

Deliverable Guidelines

The sections of greatest importance will likely be the introduction, research, project description, project hardware and software, demonstration, and conclusions. All sections of the video will likely take about 25-30 minutes to present in total. Do not exceed 35 minutes in length.

Submit your Presentation Slides (.pptx) **and** Presentation Video (.mp4) to the dropbox. If the video file is too large to submit (eg. >500MB), place it on SharePoint and submit the link too.

The dropbox will be open an additional 4 days for late submissions. For each late day your files are not submitted, this will incur a 10% reduction in marks.

Final tips:

- Write in a clear, detailed, professional tone for all sections. Provide citations.
- Support all opinions and claims with collected data evidence.
- Check your documents for: spelling, grammar, punctuation, font consistency.
- Make use of the whitespaces of your document to create comfortable spacing.
- Maintain original aspect ratios in images and crop them appropriately.
- Consider: Does your video represent the true story of your project and timeline?

Overall Marking Rubric

The following comments are provided to help guide you on producing your best quality work.

#	Criteria for Standard	3	2	1	0
	File 1: Presentation Slides (.pptx)				
	File 2: Video - exported from the PowerPoint (.mp4)				
A	Written communications: Are complete, clear, checked for errors, cited, and demonstrate a collaboration between partners. All text and visual materials are professionally presented. The <u>suspected use</u> of GenAI to produce any text or speech presented in your Presentation may result in a mark of zero for this section.				
B	References: Are included on each slide per item, at the end of the document, and match the exported video production.				
C	Key concepts: Presents relevant research, connections to recent innovations, and demonstrates thoughtful reflection on its influence and effects in the technology industry. Slide timings are professionally monitored and all work is fully cited. All project parts (MCU, IoT, inputs, outputs) are professionally identified with technical specifications that are referenced, relevant, and readable. Overall: Engaging to watch, sets a consistent tone.				
D	Material: Contents of the presentation follow instructions with all main elements requested, show evidence of editing and thoughtful progression, and is enjoyable to review.				

#	Criteria for Standard	3	2	1	0
E	Aural communications: Are clearly recorded, reduce noise in background, and demonstrate collaboration between partners. The audio overlay for each slide is relevant and concise but also develops the visuals in an engaging and informative manner.				
F	Visuals: All imagery is clear, complete, readable, and placed thoughtfully and carefully onto each page. Images are generated using CAD software. Tables are appropriately spaced, easy to read, and headings are repeated as needed across multiple pages. Students can differentiate between a system block diagram, schematic, wiring diagram, and flowchart.				
G	Engagement: All project parts (MCU, IoT, inputs, outputs) are professionally identified with technical specifications that are referenced, relevant, and readable. An advanced student would be able to follow this build. There is clear evidence of relevancy and research, references, and useful visual images. The presentation is engaging to watch and sets a consistent tone.				
H	Scientific significance: The video describes accurate and thorough details about the main topic and its natural progression into associated electronics technologies, using audio and visual components as further illustrations. The project is clearly stated.				

3 – Excellent; **2** – Satisfactory; **1** – Unsatisfactory; **0** – Poor.